

L02 — Affective Computing

Emotion theory, video features, and biosensors

Study summary generated from lecture slides. ETH AI for Digital Characters.

What is affect?

- Broader than emotion: moods, attitudes, interpersonal stances
- Three ingredients of human behavior: cognition, affect, behavior
- Applications: education (learning gain), health, digital characters

Emotion theories

- James-Lange: Stimulus -> physiological response -> emotion (sequential)
- Cannon-Bard: physiological arousal and emotion occur simultaneously
- Schachter-Singer: arousal + cognitive interpretation -> emotion
- Dimensional: valence, arousal, dominance (Russell's core affect)
- Basic emotions (Ekman): joy, sadness, disgust, surprise, anger, fear

Video-based recognition

- Pipeline: preprocessing -> feature extraction -> classification (valence/arousal)
- Action Units (FACS): 46 AUs, independent facial muscle motions
- Examples: Happiness = AU6+12; Surprise = AU1+2+5+26
- Eye blinks: AU45 per frame; Eye gaze: 9 regions
- Mouth Aspect Ratio: $MAR = (|p2-p8|+|p3-p7|+|p4-p6|) / (3*|p5-p1|)$
- Fidgeting: frame diff -> threshold -> energy % -> background update
- Classifiers: Random Forest, SVM, TCNs, Transformers, CNNs on face images

Biosensors

- ECG: electrical heart activity from skin electrodes
- PPG: light reflected by blood vessels (also via webcam)
- HRV: std of R-R intervals, RMSSD, pNN50; higher HRV -> relaxed/adaptive
- Skin conductance (GSR): sweating level, indicative of arousal
- SC decomposition: phasic (event-related) + tonic (baseline) via convex optimization (cvxEDA)
- Phasic features: amplitude, latency, rise time, half-recovery time

Artifact detection (R-R intervals)

- $QD = (Q3 - Q1) / 2$
- $MAD = (Median\ Beat - 2.9*QD) / 3$
- $MED = 3.32*QD$
- $CBD = (MAD + MED) / 2$
- Beat is artifact if $|RR - last_valid| > CBD$

Ground truth

- Self-Assessment Manikin (SAM): valence, arousal, dominance
- Other sensors: EEG, EMG, respiration